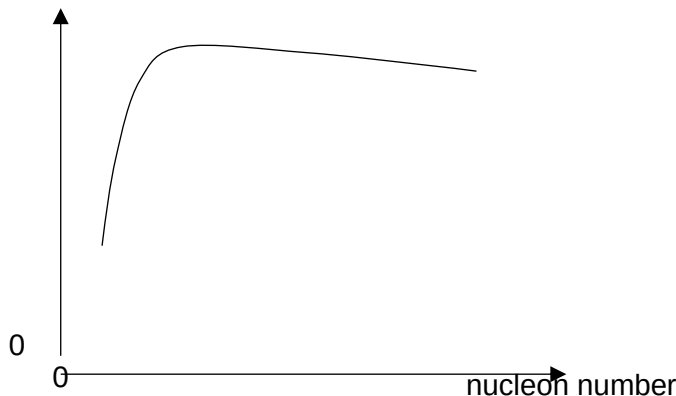


**8(a)(i)** Binding energy of a nucleus is the energy released when the nucleus is formed from its separate protons and neutrons. B1  
B1

**(a)(ii)** B1

binding energy  
per nucleon

Steeper for lower  
nucleon number



**(a)(iii)** Nuclear fusion of high nucleon number nucleus form less stable nucleus with lower binding energy per nucleon. B1

Less energy per nucleon to separate the nucleus. Therefore, energy is not released in fusion of nuclei with higher nucleon number. B1

**(a)(iv)** Neutron B1

**(a)(v)** The products/ daughter nuclei are safe, non-radioactivity nuclei. Any 1  
The reactants are readily available (from water). B1  
The energy produced per event is greater than nuclear fission.

**(a)(vi)**  $\text{Mass}_{(\text{reactants})} - \text{Mass}_{(\text{products})}$   
 $= (2.0141 \text{ u} + 3.0160 \text{ u}) - (4.0026 \text{ u} + 1.0087 \text{ u})$

[Turn over]

$$= 3.1208 \times 10^{-29} \text{ kg}$$

M1

$$\begin{aligned} E &= mc^2 \\ &= (3.1208 \times 10^{-29})(3 \times 10^8)^2 \\ &= 2.8087 \times 10^{-12} \text{ J} \end{aligned}$$

M1

$$\text{No. of reactions} = \frac{15}{5u}$$

$$\begin{aligned} \text{Total energy released} &= \left( \frac{15}{5(1.66 \times 10^{-27})} \right) (2.8086 \times 10^{-12}) \\ &= 5.1 \times 10^{15} \text{ J} \end{aligned}$$

M1

A1

**(bi)** An alpha emitter means a helium ion is released from the parent nucleus, with a reduction of 2 protons and 2 neutrons in the parent nucleus. B1

Half life is the average time taken for the number of nuclei to reduce to half the original number of nuclei. B1

Accept: Activity

**(b)(ii)**

$$\text{Activity, } |A| = \frac{dN}{dt}$$

$$I = \frac{dQ}{dt} = \frac{dN(2)(2 \times 10^5)(1.6 \times 10^{-19})}{dt}$$

M1

$$\begin{aligned} \frac{dN}{dt} &= \frac{2.4 \times 10^{-6}}{2(2 \times 10^5)(1.6 \times 10^{-19})} \\ &= 3.8 \times 10^7 \text{ Bq} \end{aligned}$$

M1

A1

**(b)(iii)** Let the number of Uranium-238 nuclei be  $N_1$   
Let the number of Uranium-235 nuclei be  $N_2$

$$N_1 = N_0 e^{-\lambda_1 t}$$

$$\frac{99.82}{100} N = N_0 e^{-(15.5 \times 10^{-11})t}$$

M1

$$N_2 = N_0 e^{-\lambda_2 t}$$

$$\frac{0.72}{100} N = N_0 e^{-(98.5 \times 10^{-11})t}$$

M1

Since  $N_0$  is the same,

$$\frac{99.82}{0.72} = e^{(-15.5 \times 10^{-11}t - (-98.5 \times 10^{-11}t))}$$

M1

$$\frac{99.82}{0.72} = e^{83 \times 10^{-11}t}$$

$$t = 5.9 \times 10^9 \text{ years}$$

A1

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